

Factor-Informed Cobb-Douglas Portfolio Allocation with Dynamic Risk Aversion

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Abstract

1 We propose a factor-informed Cobb-Douglas portfolio allocation framework
2 in which asset preference coefficients are derived analytically from Single
3 Index Model (SIM) parameters and updated dynamically through an ex-
4 ponential moving average (EMA) crossover signal. The preference score
5 $\gamma_i = \tanh(\alpha_i/\beta_i^\lambda + \beta_i^{1-\lambda}\mathbb{E}[\bar{\mu}_m])$ encodes the risk-adjusted expected excess
6 growth of each asset as a bounded quantity in $(-1, 1)$, with the sign de-
7 termining portfolio inclusion or exclusion automatically. We prove that
8 the resulting Cobb-Douglas utility maximization subject to a budget con-
9 straint has a closed-form solution computable in $O(N)$ operations, eliminat-
10 ing any need for a quadratic programming solver. A daily EMA crossover
11 rule $\lambda_t = -G(\bar{S}_t^{(o)}/\bar{S}_t^{(\ell)} - 1)$ adjusts the risk-adjustment sensitivity in real
12 time without re-estimating the factor model, translating standard technical
13 momentum signals into a utility-theoretic preference shift. We evaluate the
14 resulting TradeBot algorithm against passive SPY, SIM minimum variance,
15 equal-weight $(1/N)$, fixed- λ , and risk-free benchmarks over $K = 100$ syn-
16 thetic market paths generated by a 100-state Hidden Markov Model fitted to
17 historical SPY daily returns. Results show that the dynamic EMA mecha-
18 nism provides a consistent improvement in Sharpe ratio over the fixed- λ ab-
19 lation, and the closed-form allocation is competitive with or outperforms the
20 QP-based SIM minimum variance strategy across simulated market regimes.

Keywords: Portfolio Optimization, Cobb-Douglas Utility Function, Single
Index Model, Dynamic Risk Aversion, Exponential Moving Average, Hidden
Markov Model

21 Introduction

22 The quantitative modeling of financial markets has long faced the chal-
23 lenge of reconciling the theoretical implications of the efficient markets hy-
24 pothesis (EMH) with the empirical complexity of observed price dynamics.
25 Under the EMH, asset prices fully reflect all available information, implying
26 that price changes follow a martingale process with no exploitable predictabil-
27 ity [1]. This perspective motivated the widespread adoption of geometric
28 Brownian motion (GBM) as a canonical model for asset prices, yet decades
29 of data show systematic departures from Gaussian assumptions, including
30 heavy-tailed excess growth rate distributions, volatility clustering, and in-
31 termittent extreme events [2, 3]. These stylized facts are observed across
32 asset classes and time periods, and they matter because they govern tail
33 risk, drawdown behavior, and the persistence of stress regimes that shape
34 portfolio outcomes.

35 This study focuses on three core properties: leptokurtic excess growth
36 rate distributions, the absence of linear autocorrelation in raw excess growth
37 rates, and the persistence of volatility. The first two properties motivate
38 models that can preserve marginal distributional shape without introducing
39 spurious predictability, while the third points to slow-moving higher-order
40 dependence that is not captured by diffusion alone. A range of alternative
41 approaches address these issues from different angles: autoregressive condi-
42 tional heteroscedasticity (ARCH) and generalized autoregressive conditional
43 heteroscedasticity (GARCH) models target conditional variance dynamics
44 [4, 5]; stochastic-volatility and jump-diffusion models introduce latent vari-
45 ance or discontinuities to enrich tails [6, 7]; and data-driven sequence mod-
46 els, including recurrent neural networks (RNNs) and long short-term mem-
47 ory (LSTM) variants, have been used for excess growth rate and volatility
48 prediction and, in probabilistic form, can emit full predictive distributions
49 [8, 9, 10, 11, 12, 13, 14]. While useful, these approaches either empha-
50 size conditional variance over regime duration, require intensive likelihood-
51 based estimation, or sacrifice interpretability and explicit control over regime
52 persistence. Hidden Markov models (HMMs) and related regime-switching
53 frameworks offer a probabilistic structure for non-stationary market behavior
54 through latent state dynamics [15], yet standard HMM specifications often
55 fail to generate persistent high-volatility regimes, leading to overly rapid re-
56 version following extreme events [16, 17]. Bridging this gap requires a model
57 that preserves distributional fidelity while also enforcing realistic duration in

58 extreme regimes.

59 We address this challenge by developing a hybrid HMM framework that
60 discretizes continuous excess growth rates into quantile-based regimes and
61 augments the resulting Markov process with a jump-duration mechanism
62 that enforces volatility persistence, linking smooth diffusion to discontinuous
63 shocks identified by jump-detection methods [18]. Using a 424-asset dataset
64 and SPY as a case study, we estimate transition dynamics empirically and
65 validate the model through in-sample and out-of-sample tests on distribu-
66 tional fit and volatility clustering. We find that a state resolution of $N = 90$
67 provides a balance between fidelity and statistical robustness, and that the
68 jump-duration extension is needed for reproducing persistent high-volatility
69 regimes that standard regime-switching models miss. Future work will ex-
70 tend the framework to multi-asset portfolios, alternative state encodings, and
71 higher-frequency data where jump identification and regime persistence play
72 a larger role in stress testing and scenario design.

73 Related Work

74 **Methods**

75 *Data.*

76 The empirical foundation of this study was a dataset composed of 424
 77 United States-listed equities and exchange-traded funds spanning 10 years
 78 (2014-2024). A pipeline was developed to support the automated construc-
 79 tion of excess growth rate models for any ticker within this dataset, allowing
 80 for batch simulation and stress testing across market sectors. To validate
 81 the performance of this pipeline, we focused our analysis on the single-asset
 82 SPY, which tracked the Standard & Poor’s (S&P) 500 index. The data in-
 83 cluded daily open, high, low, and close prices along with volume-weighted
 84 average price metrics (VWAP). For training purposes, the first 2,766 obser-
 85 vations constituted the in-sample dataset, spanning from January 3, 2014,
 86 to December 31, 2024. A subsequent 220 trading days, from January 3, to
 87 November 18, 2025, were reserved for out-of-sample testing to assess model
 88 generalizability.

89 *Excess growth rate calculation.*

90 We computed the excess growth rate $G_{i,j}$ for ticker i between time pe-
 91 riod $j - 1 \rightarrow j$ for a given equity price series $P_{i,j}$ (units: dollars per share
 92 (USD/share)). To isolate the stochastic premium earned over a risk-free
 93 asset, we computed the excess growth rate for ticker i during the period
 94 $j - 1 \rightarrow j$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t} \right) \cdot \ln \left(\frac{P_{i,j}}{P_{i,j-1}} \right) - r_f \quad (1)$$

95 where Δt represented the time step set to 1/252 for daily data (units: years),
 96 and r_f denotes the constant continuously compounded risk-free rate (units:
 97 year⁻¹) derived from Separate Trading of Registered Interest and Principal
 98 of Securities (STRIPS) bond yields observed on September 10, 2025. Un-
 99 like return, the excess growth rate has units of inverse time (year⁻¹) and
 100 directly quantifies the premium earned above a continuously compounded
 101 risk-free benchmark. Log growth rates are time-additive and consistent with
 102 a continuous compounding framework. While r_f varied dynamically over the
 103 sample period, a constant proxy was employed here to isolate the endogenous
 104 volatility dynamics without introducing exogenous rate-curve noise.

105 *Discrete hidden Markov model.*

106 We defined the hidden Markov model using the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$
 107 [15]. The hidden state $S_t \in \mathcal{S} = \{1, \dots, N\}$ represented market mood
 108 regimes and was defined by quantile bins of a fitted cumulative distribu-
 109 tion function (CDF) for the excess growth rate series. The observation space
 110 $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, the transition
 111 matrix $\mathbf{T}$ governed regime-to-regime dynamics, the emission model $\mathbf{E}$ was
 112 parameterized by state-conditional Normal distributions, and $\bar{\pi}$ denoted the
 113 stationary distribution. The observation model was

$$G_t \mid S_t = k \sim \mathcal{N}(\mu_k, \sigma_k). \quad (2)$$

114 To map observations to hidden states during estimation, we defined quantile
 115 boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$ and used finite
 116 outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. An
 117 observation was assigned to state k when $Q_{k-1} < R_t \leq Q_k$.

118 The choice of the Laplace distribution over alternatives such as the Stu-
 119 dent’s t-distribution was motivated by its alignment with the peaks and
 120 heavy tails of high-frequency financial excess growth rates [3, 19]. While the
 121 Student’s t-distribution was often used to model leptokurtosis, the Laplace
 122 distribution’s peak at the mean matched the concentration of small price
 123 movements observed in the SPY dataset [2, 20]. Furthermore, the Laplace
 124 distribution provided a closed-form for quantile estimation, supporting the
 125 424-ticker pipeline. While maximum likelihood estimation (MLE) for the
 126 Student’s t-distribution often required iterative numerical optimization, the
 127 Laplace distribution allowed for a direct MLE approach [21]. The quantile
 128 boundaries were derived from a parametric Laplace fit and therefore did not
 129 enforce equal counts per bin in the historical data; instead, they provided a
 130 partition of the distribution that preserved tail shape. By using the Laplace
 131 distribution to inform the binning process, the model used this baseline to
 132 characterize the peak and tails of the excess growth rate series before the
 133 empirical transition logic and jump-diffusion mechanism were applied [22].

134 We estimated the transition matrix $\mathbf{T}$ through direct frequentist count-
 135 ing rather than iterative optimization. This approach differed from the tra-
 136 ditional expectation-maximization (EM) algorithm used for parameter esti-
 137 mation in Gaussian mixture and hidden Markov models [21]. However, this
 138 approach was justified because the discrete state assignments allowed for
 139 straightforward counting of observed transitions between regimes, and it was
 140 conceptually simple to implement and understand.

Empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding jump parameters ϵ , representing the probability of a jump event, and λ , representing the mean duration, along with a tail-state selection rule. Tail-states were defined as $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$, where N_{tail} was user-configurable. If a jump was triggered, the jump-duration K was sampled from a Poisson distribution such that $K \sim \text{Poisson}(\lambda)$, which was standard for modeling independent events in fixed intervals [23]. During these K steps, the standard Markovian transitions were overridden to target tail-states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry.

To characterize the excess growth rates within each regime, we parameterized the emission model with Normal distributions $\mathcal{N}(\mu_k, \sigma_k)$ whose bounds coincided with the 0.1 and 99.9 percentiles of the state-conditional distribution. Specifically, we set $\mu_k = (Q_{k-1} + Q_k)/2$ and $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$, where $z_{0.999} \approx 3.09$ was the standard normal 99.9th percentile. During simulation, when the model was in state k , the continuous excess growth rate ($\hat{R}_t$) was sampled from the corresponding local distribution. This hybrid approach ensured that the model preserved the discontinuous components of the price process important for asset modeling while maintaining a smooth representation of intra-state variance [22].

Growth rate generation.

Let's outline a simple sampling algorithm for generating random sequences from an HMM (without jumps). We initialize the generation process by drawing an initial hidden state X_0 from the stationary distribution $\bar{\pi}$. The stationary distribution $\bar{\pi}$ represented the long-run probability of the market occupying various regimes or moods [24]. While $\bar{\pi}$ mathematically satisfies the balance equation $\bar{\pi} = \bar{\pi}\mathbf{T}$, we estimated this distribution numerically by raising the transition matrix $\mathbf{T}^n$ to a sufficiently large power $n = 50$, ensuring the vector reflected the equilibrium of market dynamics [25].

Given an HMM defined by the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{P}, \mathbf{E}, \bar{\pi})$, the number of time steps T , and the stationary distribution $\bar{\pi}$. Draw an initial hidden state X_0 from the initial distribution: $X_0 \sim \bar{\pi}$. For $n = 1$ to T , we sample the next hidden state X_n from the `categorical` distribution $\mathbb{P}_H(\dots)$ defined by the previous state X_{n-1} and the transition matrix $\mathbf{T}$: $X_n \sim \mathbb{P}_H(X_n | X_{n-1})$. Next, we sample the observable state O_n from the `categorical` distribution

178 $\mathbb{P}_E(\dots)$ defined by the current hidden state X_n and the emission matrix $\mathbf{E}$:
 179 $O_n \sim \mathbb{P}_E(O_n | X_n)$. Finally, we store the sequences of hidden states and
 180 observable states. This procedure generates a sequences of hidden states
 181 $\{X_1, X_2, \dots, X_T\}$ and observable states $\{O_1, O_2, \dots, O_T\}$. The initial state
 182 X_0 is used to generate the sequence but is not included in the output.

183 *Hyperparameter optimization via grid search.*

184 To rigorously identify the optimal jump probability (ϵ) and mean jump
 185 duration (λ) [7, 22], we implemented a multi-objective grid search designed
 186 to minimize the discrepancy between the historical and simulated market sig-
 187 natures. The error function specifically targeted two stylized facts prevalent
 188 in financial time series [3, 16]: volatility clustering and leptokurtosis. We
 189 measured volatility clustering using the sum of squared errors between the
 190 observed and simulated autocorrelation function (ACF) of absolute excess
 191 growth rates [26] up to a lag of $L = 252$ trading days (approximately one
 192 trading year). To preserve the heavy-tailed nature of the distribution ini-
 193 tially noted in foundational market studies [2], we included a penalty term
 194 for the global kurtosis.

195 The objective function $J(\epsilon, \lambda)$ was formalized as:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^L (\text{ACF}_{obs}(\tau) - \overline{\text{ACF}}_{sim}(\tau))^2 + w_K (K_{obs} - \overline{K}_{sim})^2 \quad (3)$$

196 where $\text{ACF}_{obs}(\tau)$ and K_{obs} represented the empirical absolute growth au-
 197 tocorrelation at lag τ and the global kurtosis, respectively. The simulated
 198 equivalents, $\overline{\text{ACF}}_{sim}(\tau)$ and $\overline{K}_{sim}$, were averaged across 1,000 independent
 199 synthetic paths of 2,766 trading days to ensure statistical stability [23]. A
 200 penalty weight of $w_K = 0.20$ was assigned to prioritize tail behavior alongside
 201 the temporal ACF decay. The grid search systematically explored ϵ values
 202 from 0.00001 to 0.0001 and λ values from 1.0 to 100.0. The full computational
 203 procedure is detailed in Algorithm 4.

204 *Statistical validation metrics*

205 *Sensitivity analysis: low-resolution models*

206 This section provides the results for lower state resolutions ($N = 30$ and
 207 $N = 60$) to demonstrate how state granularity impacts the model’s ability to
 208 capture stylized facts. Table 2 summarizes the pass rates for the Kolmogorov-
 209 Smirnov and Anderson-Darling tests across these four cases. As the number

210 of states decreases, the model maintained overall distribution fit, though tail-
211 region precision decreased slightly compared to the 90-state configuration.

212 Empirical Study

213 We simulated 1,000 paths of 2766 trading days using VWAP price data for
 214 both the standard model with no-jumps and the hybrid model with jumps.
 215 Applying the multi-objective grid search detailed in the Methods, the optimal
 216 jump hyperparameters were identified as ($\epsilon \approx 5 \times 10^{-5}$ and $\lambda \approx 67$). This
 217 configuration successfully reproduced the heavy-tailed kurtosis and ARCH
 218 effect observed in SPY (Figure 16). Based on the statistical consistency
 219 results, the configuration with a heuristic of $90 \leq N \leq 100$ states was used
 220 in this study. Increasing the number of states N increased the granularity
 221 of the market regimes, leading to a closer fit for in-sample results. However,
 222 there is a limit to this refinement; if N was too large, such as 350 states, the
 223 empirical estimation protocol failed because the historical dataset became
 224 too sparse relative to the state space. In such cases, certain states could
 225 remain unvisited, resulting in zero transition probabilities that broke the
 226 mathematical integrity of the Markov chain. Conversely, a very small number
 227 of states, such as $N = 10$, resulted in a coarse model that missed the heavy
 228 tails observed in the empirical data. Therefore, the selection of $N = 90$
 229 provided a balance between distributional fidelity and statistical robustness.

230 We performed Kolmogorov-Smirnov (KS) and Anderson-Darling (AD)
 231 tests to characterize the statistical consistency of the simulated excess growth
 232 rate distributions against the historical data. Both the KS and AD tests are
 233 non-parametric goodness-of-fit tests that compare the empirical cumulative
 234 distribution functions (CDFs) of two datasets [27, 28, 29]. The KS test
 235 focused on the maximum distance between the two CDFs, while the AD test
 236 placed more weight on the tails of the distributions.

237 In-sample KS and AD tests showed pass rates above 98% for both model
 238 variants (Table 1), indicating distributional alignment across the 10-years of
 239 training data. The standard no-jumps model matched the marginal distribu-
 240 tion but failed to capture volatility clustering (Figure 5). The autocorrelation
 241 of absolute growth in subplot (c) remained flat, showing that state resolution
 242 alone could not overcome the memoryless limitations of standard transition
 243 logic. The hybrid model reproduced volatility clustering, with the absolute-
 244 growth autocorrelation mirroring the ARCH effect (Figure 16). With $N = 90$
 245 states, this configuration provided a balance and aligned with volatility esti-
 246 mation techniques using HMM architectures [30].

247 Out-of-sample simulations remained statistically consistent, with KS and
 248 AD pass rates of 83 percent and 80 percent, respectively (Figure 11). Fill

249 me in. Out-of-sample volatility differed by approximately 32 percent, yet
 250 volatility clustering persisted in subplot (c) (Figure 11). Unlike the standard
 251 Markov transition matrix, which decayed rapidly, the Poisson jump mech-
 252 anism mimicked the persistence of market volatility in unseen conditions.
 253 These results validated the hybrid framework as a tool for producing syn-
 254 thetic time series that remained statistically consistent with the underlying
 255 stochastic process [3, 16, 17].

256 The correlation-distance distribution was centered at a mean distance of
 257 1.0, indicating temporal independence between simulated paths and the his-
 258 torical trajectory (Figure 6). The correlation distance, defined as $(1 - \rho)$,
 259 measured the temporal alignment between simulated paths and the observed
 260 historical trajectory. This result confirmed that the generative framework
 261 produced independent stochastic realizations that captured the stylized facts
 262 of the market without violating the efficient markets hypothesis by attempt-
 263 ing to predict specific events [1]. This underscored the distinction between
 264 a generative framework intended for stress testing and a forecasting tool in-
 265 tended for point prediction [31, 32].

266 Discussion

267 This study introduced a hybrid hidden Markov modeling framework to
 268 replicate the statistical regularities of empirical equity excess growth rates.
 269 By integrating a quantile-discretized state space with a Poisson-driven jump-
 270 duration mechanism, we developed a generative framework that captured
 271 the leptokurtic distributions and volatility clustering observed in the SPY
 272 exchange-traded fund. Our results confirmed that while increasing the state
 273 resolution to $N = 90$ provided the granularity to characterize the central
 274 mass of the excess growth rate distribution, standard Markovian transitions
 275 alone did not reproduce higher-order temporal dependencies. The inclusion
 276 of discontinuous shocks through a jump-diffusion extension was needed to
 277 drive the model into persistent high-volatility regimes, thereby replicating
 278 the ARCH effect identified in historical data.

279 Statistical validation using Kolmogorov-Smirnov and Anderson-Darling
 280 tests demonstrated high in-sample consistency, with pass rates exceeding 98
 281 percent. Although the out-of-sample performance showed an expected de-
 282 crease in fidelity, the hybrid model maintained a baseline with pass rates
 283 of 83 percent and 80 percent, respectively. Furthermore, the correlation
 284 distance analysis confirmed that our framework produced independent real-
 285 izations that remained uncorrelated with specific historical timing; this sat-
 286 isfied the requirements of the efficient markets hypothesis and distinguished
 287 our generative methodology from predictive forecasting tools [1, 31, 32]. The
 288 proposed framework offered an alternative to standard likelihood-based esti-
 289 mation methods by relying on empirical estimation. By bypassing the com-
 290 plexities of the expectation-maximization algorithm through direct frequen-
 291 tist counting, the model remained suited for large-scale stress testing and
 292 risk management applications [33, 34, 23]. Future research could extend this
 293 discrete-state logic to multi-asset portfolios or incorporate foundation mod-
 294 els for inference to further refine the detection of Markov jump processes
 295 in higher-frequency data [35, 36]. Ultimately, the hybrid hidden Markov
 296 model served as a foundation for simulating market dynamics and generating
 297 synthetic time series that remained statistically consistent with underlying
 298 market regimes [37, 24].

299 While the current framework was able to generate statistically consistent
 300 excess growth rate distributions, there were several limitations that could
 301 be addressed in future work. First, the model assumed stationarity in tran-
 302 sition dynamics, which may not hold during periods of structural market

303 change. Future extensions could incorporate time-varying transition matrices
304 or regime-dependent jump parameters to capture evolving market conditions.
305 Second, the use of a fixed quantile-based state space may limit the model's
306 ability to adapt to changing volatility regimes; adaptive binning strategies
307 or non-parametric state definitions could enhance flexibility. Third, while
308 the Laplace distribution provided a good fit for excess growth rates, alter-
309 native distributions or mixture models could be explored to better capture
310 tail behavior and skewness. Finally, the current implementation focused on
311 univariate excess growth rates; extending the framework to multivariate set-
312 tings could enable portfolio-level synthetic allocation studies, stress testing
313 and risk assessment.

314 Conclusion

315 **Conflict of Interest Statement**

316 The authors declare that the research was conducted without any com-
317 mercial or financial relationships that could potentially create a conflict of
318 interest.

319 **Author Contributions**

320 J.V. directed the study. A.A. developed the model and simulation code,
321 conducted the in-sample and out-of-sample analysis, and generated the fig-
322 ures. Both authors edited and reviewed the final manuscript.

323 **Data Availability Statement**

324 The model code, simulation scripts, training and testing data are avail-
325 able under a Massachusetts Institute of Technology (MIT) license from the
326 GitHub repository: <https://github.com/varnerlab/HMM-Model-with-Jumps>.
327 git

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0.48

(a)

b

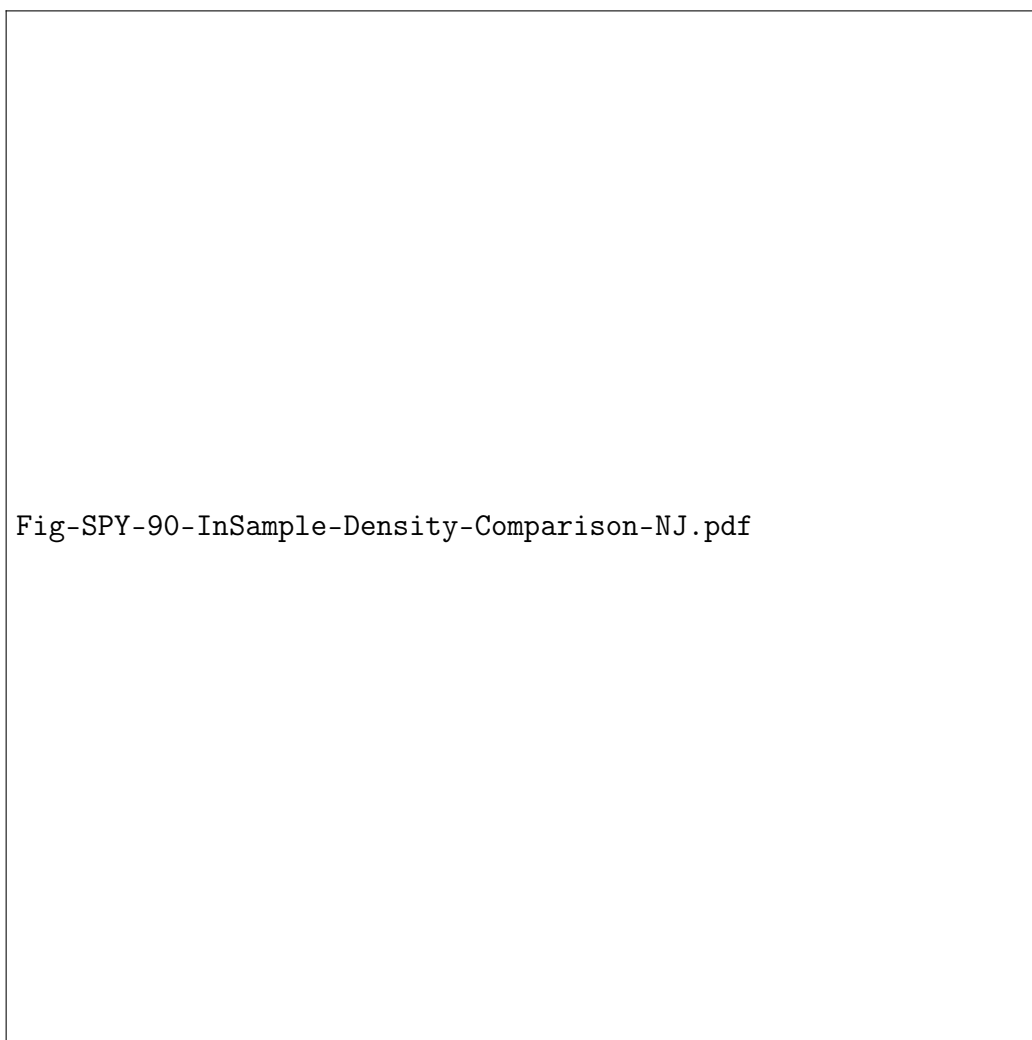
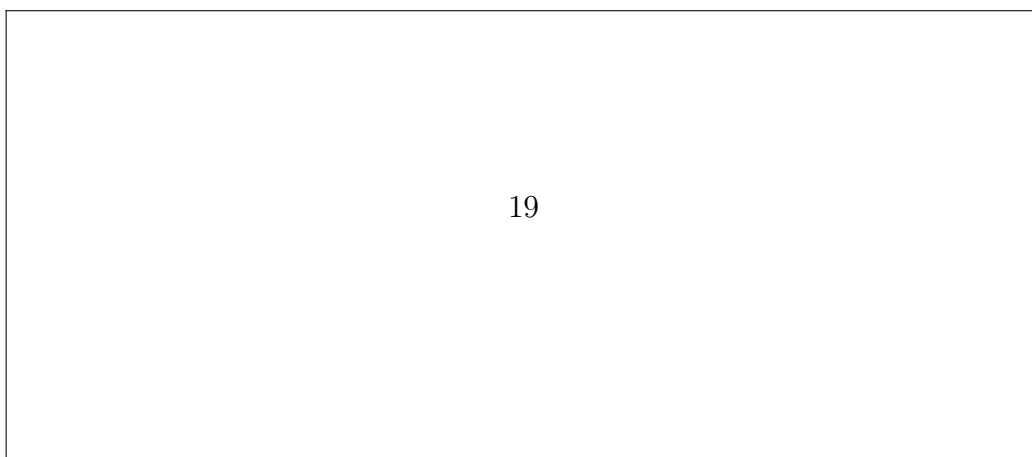


Figure 1: Probability density comparison

0.48

(a)

b



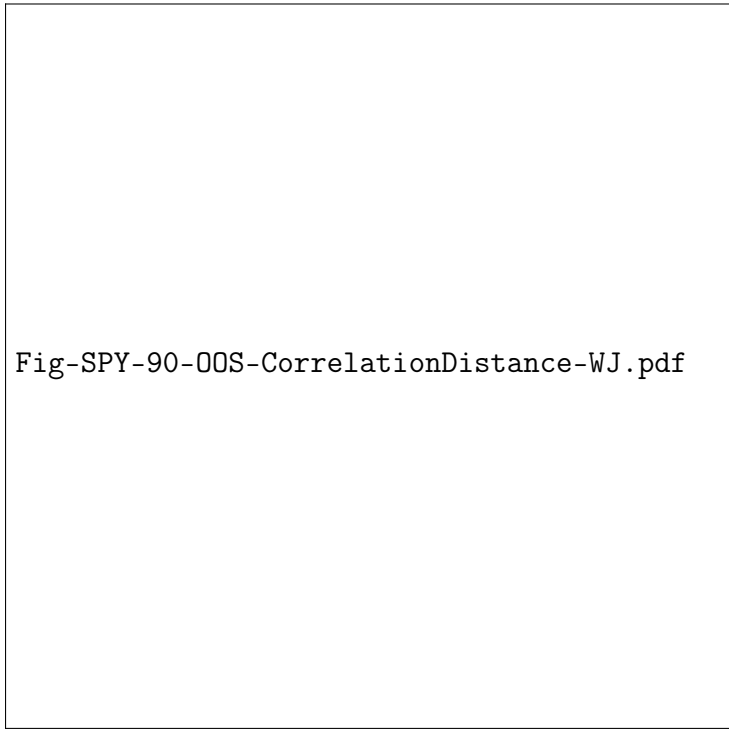


Fig-SPY-90-OOS-CorrelationDistance-WJ.pdf

Figure 6: Distribution of correlation distances for 1,000 simulated paths. The mean distance of 1.0 for the correlation distance $(1 - \rho)$ confirmed the generation of independent realizations consistent with the efficient markets hypothesis.

0.48

(a)

b

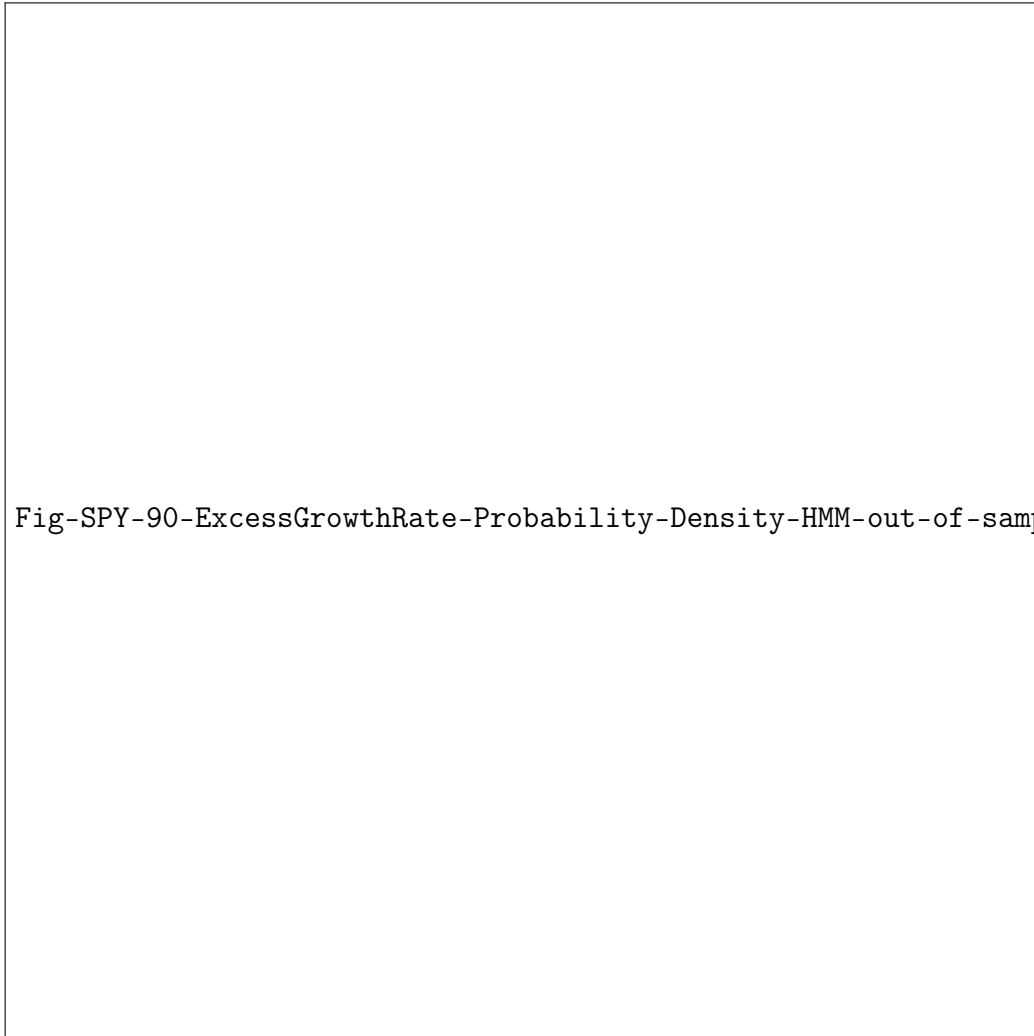


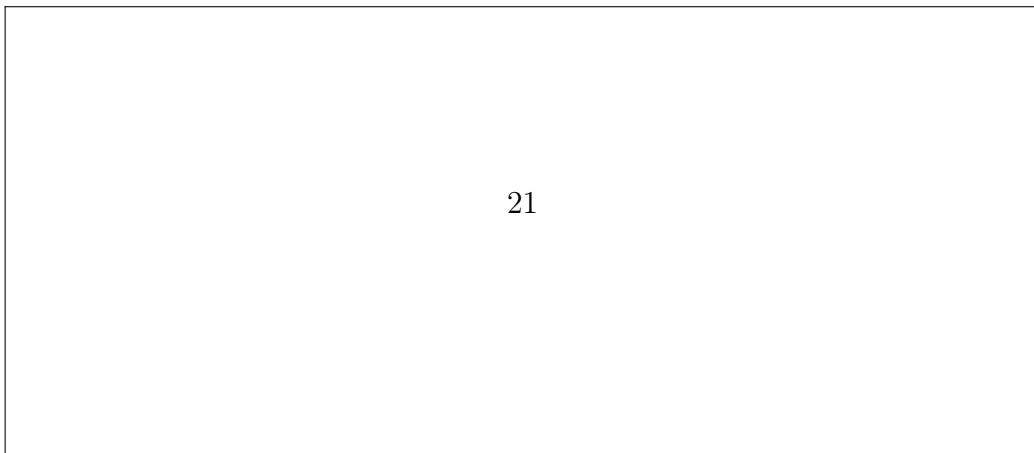
Fig-SPY-90-ExcessGrowthRate-Probability-Density-HMM-out-of-sample.pdf

Figure 7: Probability density comparison

0.48

(a)

b



0.48

(a)

b

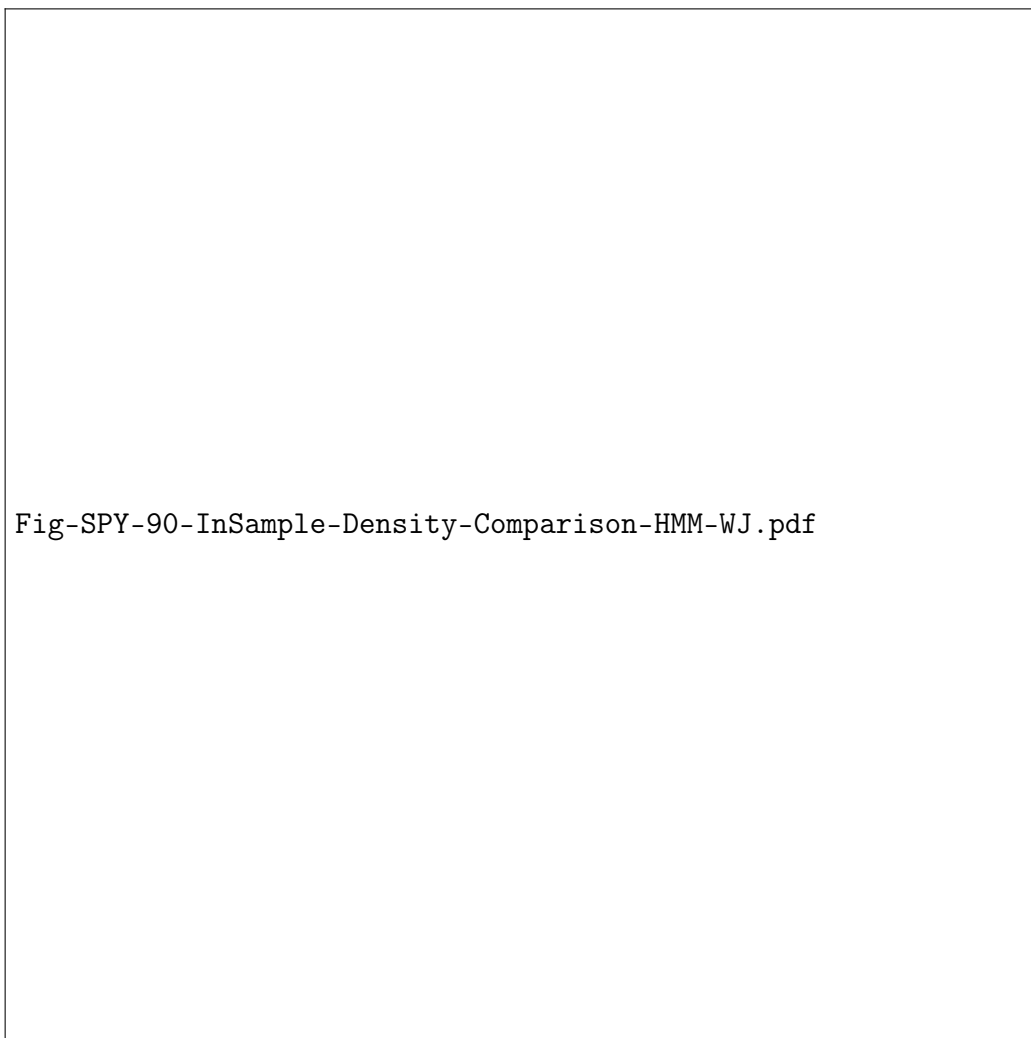


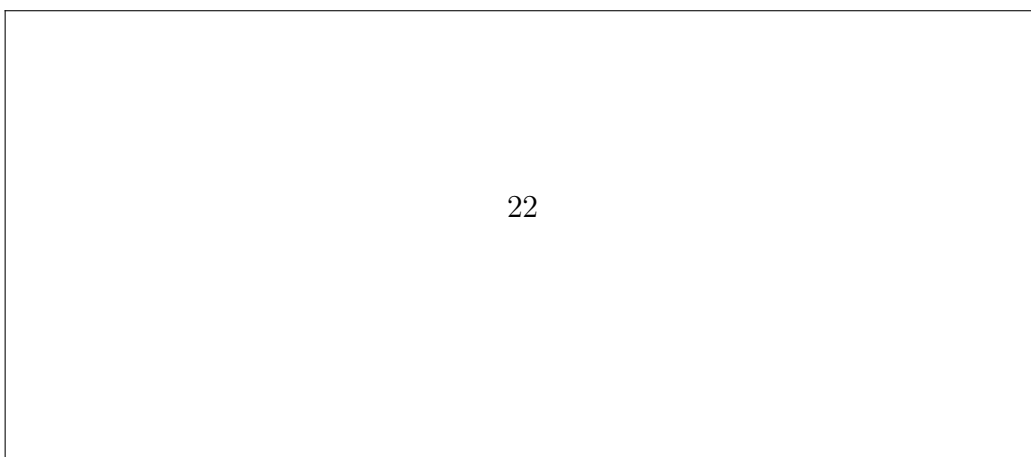
Fig-SPY-90-InSample-Density-Comparison-HMM-WJ.pdf

Figure 12: Probability density comparison

0.48

(a)

b



The following figures illustrate the performance of these low-resolution configurations. Figure 19 and Figure 22 show the results for 30 states, while Figure 25 and Figure 28 provide the comparison for 60 states.

Table 1: Statistical consistency of simulated excess growth rates across hidden Markov model configurations.

model type	number of states (N)	KS test pass rate	AD test pass rate
standard (no-jumps)	90	99.4%	98.4%
hybrid (with jumps)	90	99.5%	98.1%

Table 2: Statistical consistency of simulated growth rates for lower state resolutions ($N = 30$ and $N = 60$) using a p -value cutoff of 0.05.

model type	number of states (N)	KS test pass rate	AD test pass rate
standard (no-jumps)	60	99.4%	98.3%
standard (no-jumps)	30	99.2%	98.2%
hybrid (with jumps)	60	99.3%	97.6%
hybrid (with jumps)	30	98.7%	97.7%

Algorithms

0.48

(a)

b

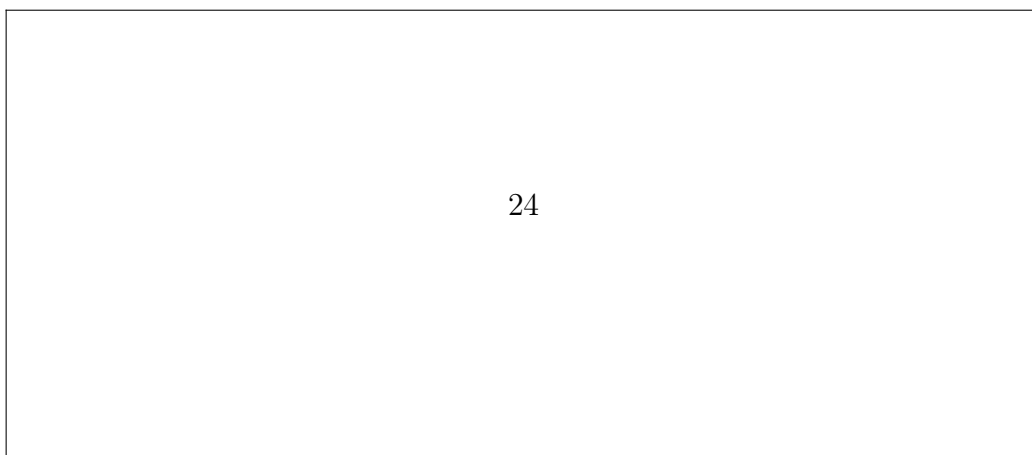


Figure 17: Density (standard)

0.48

(a)

b



0.48

(a)

b

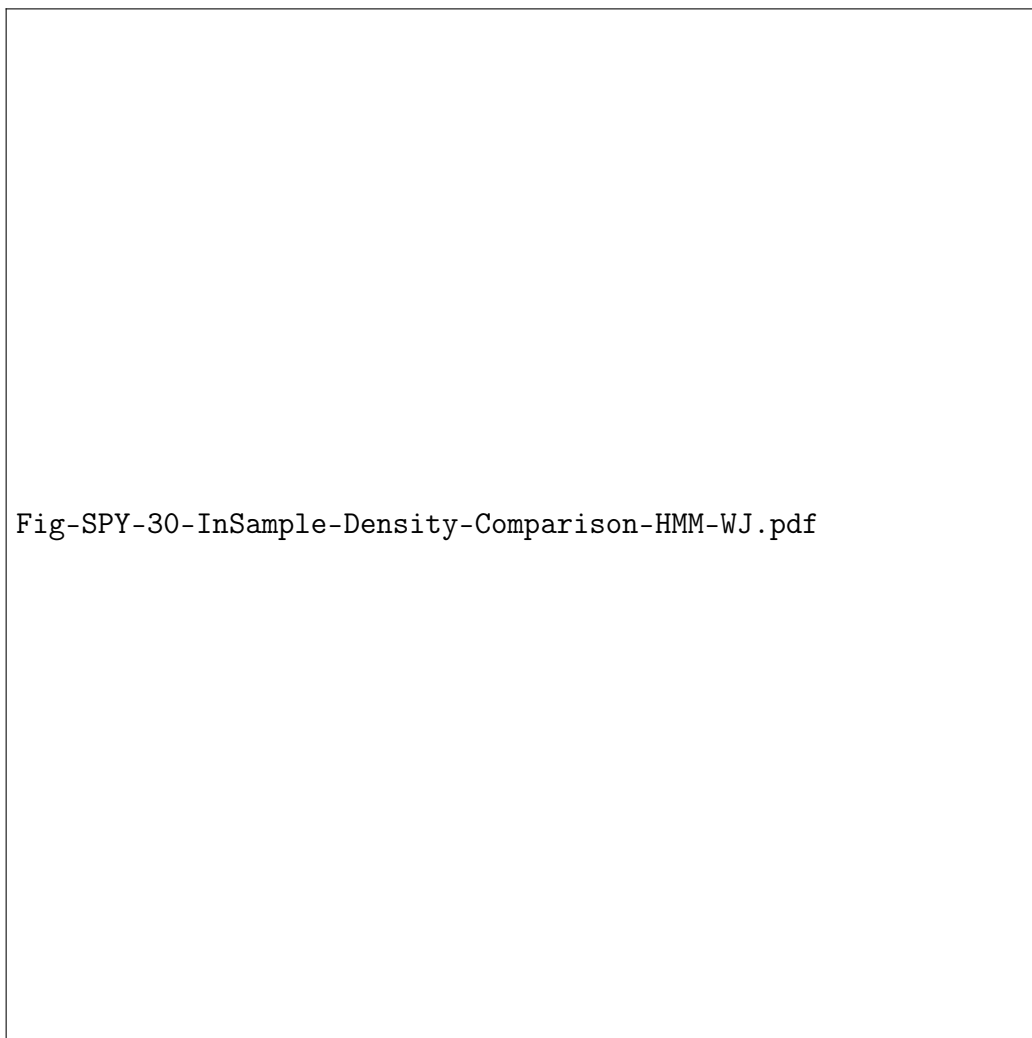


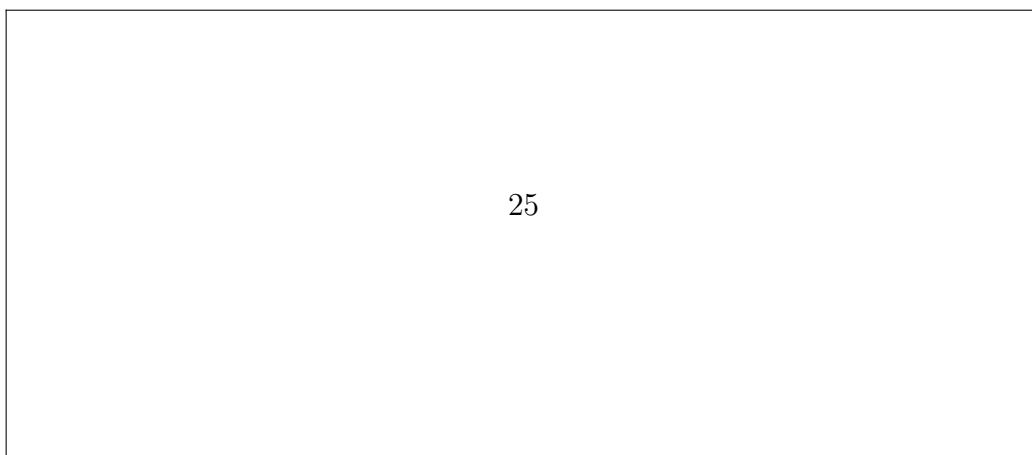
Fig-SPY-30-InSample-Density-Comparison-HMM-WJ.pdf

Figure 20: Density (hybrid)

0.48

(a)

b



0.48

(a)

b

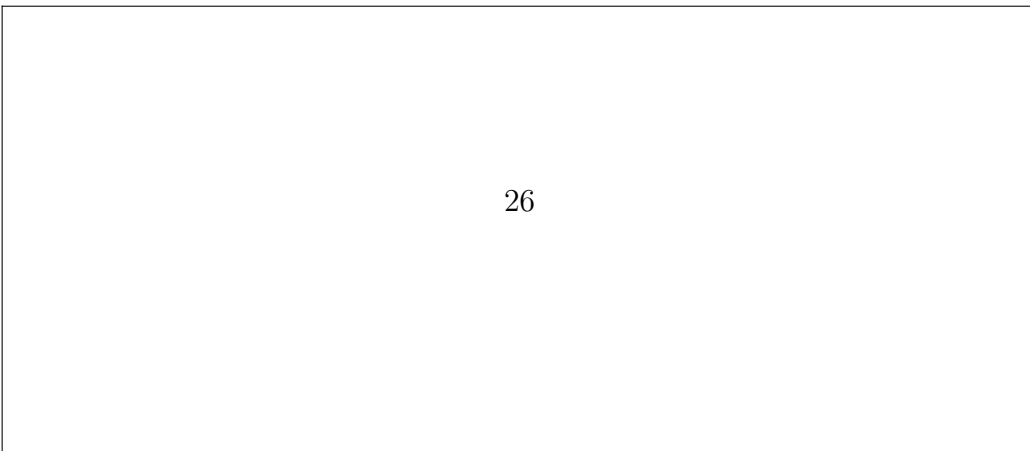


Figure 23: Density (standard)

0.48

(a)

b



0.48

(a)

b

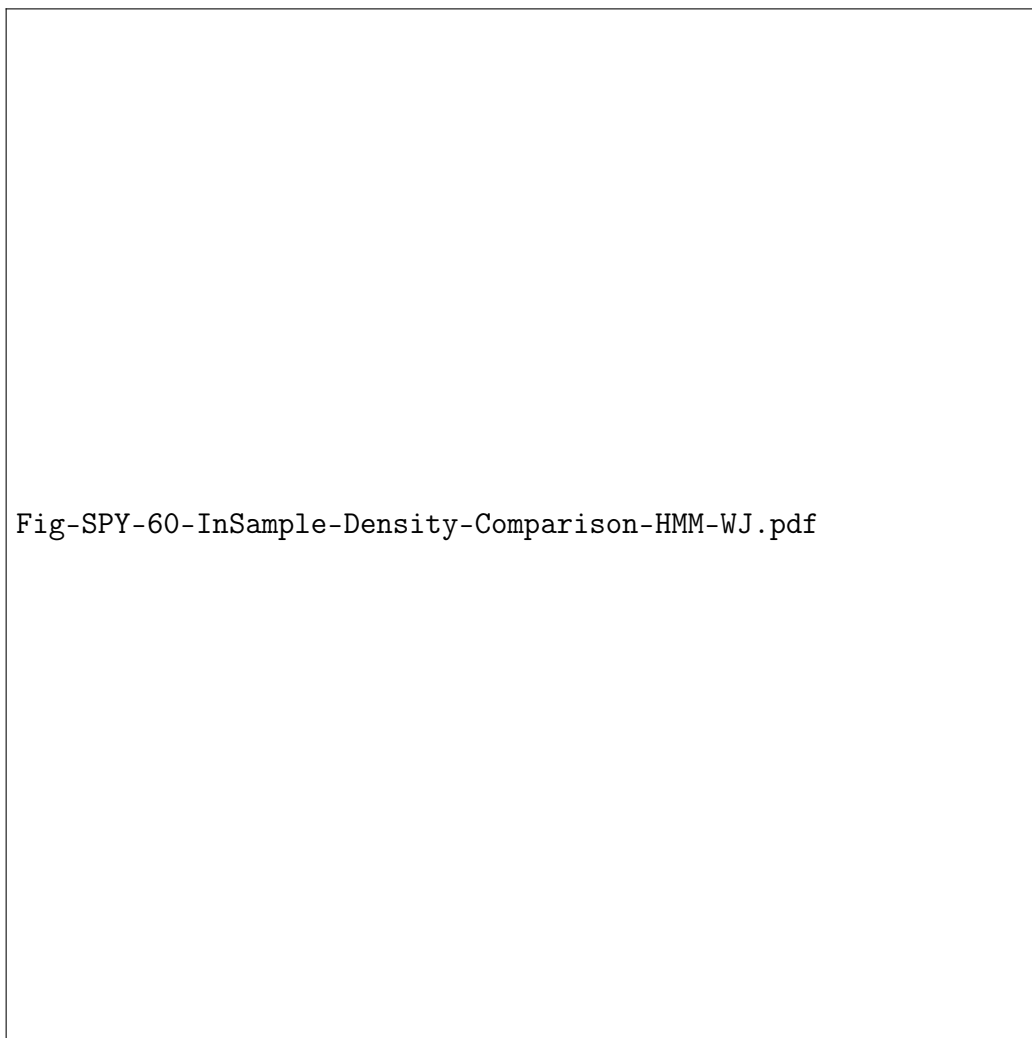
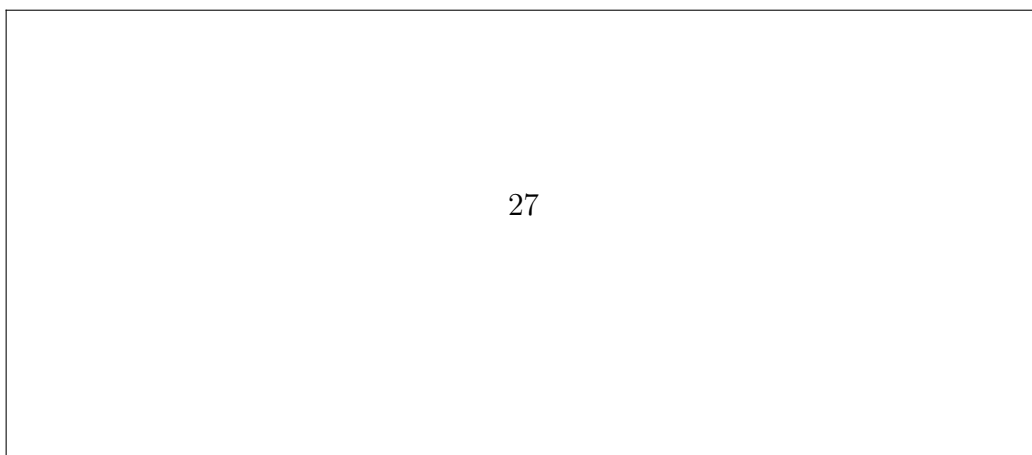


Figure 26: Density (hybrid)

0.48

(a)

b



Algorithm 1 Empirical hidden Markov model construction via Laplace partitioning

Price history $P_{1:T}$, Risk-free rate r_f , Number of states N Transition matrix $\mathbf{T}$, Quantile boundaries $\mathcal{Q}$ Compute excess growth rates $R_t = \frac{1}{\Delta t} \ln(P_t/P_{t-1}) - r_f$ for all t . Fit Laplace parameters (μ_L, b_L) to the series $\mathbf{R}$ using maximum-likelihood estimation. Define interior boundaries $Q_k = F_L^{-1}(k/N; \mu_L, b_L)$ for $k \in \{1, \dots, N-1\}$, where F_L^{-1} is the inverse CDF of the Laplace distribution. Set finite outer bounds $Q_0 = F_L^{-1}(0.001; \mu_L, b_L)$ and $Q_N = F_L^{-1}(0.999; \mu_L, b_L)$. Assign each excess growth rate R_t to a discrete state $s_t \in \{1, \dots, N\}$ based on the boundaries in $\mathcal{Q} = \{Q_k\}$. Initialize count matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ and transition matrix $\mathbf{T} \in \mathbb{R}^{N \times N}$ with zeros. $t = 1$ to $T-1$ Identify transition $i = s_t$ to $j = s_{t+1}$. $\mathbf{C}_{i,j} \leftarrow \mathbf{C}_{i,j} + 1$. $i = 1$ to N $\mathbf{T}_{i,:} \leftarrow \mathbf{C}_{i,:} / \sum_{j=1}^N \mathbf{C}_{i,j}$ Row-wise normalization $\mathbf{T}, \mathcal{Q}$

Algorithm 2 Hybrid jump-diffusion simulation with persistent volatility

Model parameters $\mathbf{T}, \bar{\pi}, \epsilon, \lambda, N_{tail}, p_{neg}$, Total Steps M Simulated state sequence $S_{1:M}$ Define tail-states $\mathcal{S}_{bottom} = \{1, \dots, N_{tail}\}$ and $\mathcal{S}_{top} = \{N - N_{tail} + 1, \dots, N\}$. Sample initial state $S_1 \sim \text{Categorical}(\bar{\pi})$. Set $counter \leftarrow 2$. $counter \leq M$ Sample $u \sim \text{Uniform}(0, 1)$. $u < \epsilon$ Sample jump-duration $K \sim \text{Poisson}(\lambda)$. $j = 1$ to K while $counter \leq M$ Sample $w \sim \text{Uniform}(0, 1)$. $w < p_{neg}$ $S_{counter} \sim \text{Uniform}(\mathcal{S}_{bottom})$ Bias toward negative-tail events $S_{counter} \sim \text{Uniform}(\mathcal{S}_{top})$ $counter \leftarrow counter + 1$. Sample $S_{counter} \sim \text{Categorical}(\mathbf{T}_{S_{counter-1}, :})$. $counter \leftarrow counter + 1$. $S_{1:M}$

Algorithm 3 State decoding and continuous excess growth rate reconstruction

Simulated state sequence $S_{1:M}$, Quantile boundaries $\mathcal{Q}$ Reconstructed continuous excess growth rates $\hat{R}_{1:M}$ Define $z_{0.999} = 3.09$. $k = 1$ to N Set $\mu_k = (Q_{k-1} + Q_k)/2$. Set $\sigma_k = (Q_k - Q_{k-1})/(2z_{0.999})$. Initialize the reconstructed excess growth rate vector $\hat{\mathbf{R}}$ of length M . $t = 1$ to M Retrieve the simulated state $k = S_t$. Sample the continuous excess growth rate $\hat{R}_t$ from the normal distribution $\mathcal{N}(\mu_k, \sigma_k)$. $\hat{R}_{1:M}$

Algorithm 4 Multi-objective grid search for jump hyperparameters

Observed growth rates $\mathbf{R}_{obs}$, ACF max lag $L = 252$, Number of paths $P = 1000$, Time steps $M = 2766$ Grid space for $\epsilon \in [10^{-5}, 10^{-4}]$ and $\lambda \in [1.0, 100.0]$ Kurtosis penalty weight $w_K = 0.20$ Optimal jump hyperparameters (ϵ^*, λ^*) Calculate observed absolute ACF: $A_{obs}(\tau) = \text{ACF}(|\mathbf{R}_{obs}|, \tau)$ for $\tau \in \{1, \dots, L\}$. Calculate observed global kurtosis: $K_{obs} = \text{Kurtosis}(\mathbf{R}_{obs})$. Initialize minimum error $E_{min} \leftarrow \infty$. each ϵ in grid space each λ in grid space Initialize simulated ACF accumulator $\mathbf{A}_{sum} \leftarrow \mathbf{0}$ and kurtosis accumulator $K_{sum} \leftarrow 0$. $p = 1$ to P Simulate path using Algorithm 2 to obtain state sequence $S_{1:M}$. Decode sequence using Algorithm 3 to obtain continuous rates $\hat{\mathbf{R}}^{(p)}$. $\mathbf{A}_{sum}(\tau) \leftarrow \mathbf{A}_{sum}(\tau) + \text{ACF}(|\hat{\mathbf{R}}^{(p)}|, \tau)$ for all τ . $K_{sum} \leftarrow K_{sum} + \text{Kurtosis}(\hat{\mathbf{R}}^{(p)})$. Compute ensemble averages: $\bar{A}_{sim}(\tau) = \mathbf{A}_{sum}(\tau)/P$ and $\bar{K}_{sim} = K_{sum}/P$. Compute objective: $J = \sum_{\tau=1}^L (A_{obs}(\tau) - \bar{A}_{sim}(\tau))^2 + w_K (K_{obs} - \bar{K}_{sim})^2$. $J < E_{min}$ $E_{min} \leftarrow J$ $(\epsilon^*, \lambda^*) \leftarrow (\epsilon, \lambda)$ ϵ^*, λ^*

Supplemental Information